LESSON 3.4

Determine Equivalent Ratios Using Graphs

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can graph the values from a ratio table as points on the coordinate plane.

WHY THIS MATTERS The organizers of a soccer league ordered 6 of the bags shown to hold soccer balls. Each bag holds the same number of soccer balls. How many soccer balls will the bags hold?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Coordinate plane *Spanish: Plano cartesiano*	A grid with a line going across and a line going up to plot points.	x across, y up
Ordered pair *Spanish: Par ordenado*	Two numbers (x, y) that tell where a point is on a grid.	(2, 14)
Linear pattern *Spanish: Patrón lineal*	Points that make a straight line on a grid.	(1,2), (2,4), (3,6) all line up
Proportional *Spanish: Proporcional*	Two amounts that grow together at the same rate.	A straight line from (0,0) through (2,6) and (4,12)
Origin *Spanish: Origen*	The point (0, 0) where the two grid lines cross.	0 pages, 0 stickers
Ratio table *Spanish: Tabla de razones*	A table of equal ratios that shows a pattern.	Cups of juice: 2, 4, 6 | Cups of water: 3, 6, 9

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Each tray holds 8 muffins (x = trays, y = muffins). Write and plot three points.

1. Table: 1 tray = 8, 2 trays = 16, 3 trays = 24.
2. Pairs: (1, 8), (2, 16), (3, 24).
3. Plot each pair from (0, 0): across x, then up y.

Answer: The points lie on one straight line through (0, 0)

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

A recipe uses 1 cup of sugar for every 3 cups of flour.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The point (_____ , _____) means _____ .
- The origin means _____ .
- I found y by multiplying x by _____ .
- The points form a straight line because _____ .

Word bank coordinate plane · ordered pair · origin · x-axis · y-axis · plot · point · straight line · table · multiply · across · up

WATCH OUT!

A common mistake

A common mistake in Graph Ratio Tables is writing the ordered pair in the wrong order

LESSON 3.4

Determine Equivalent Ratios Using Graphs

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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LEARNING TARGET I can graph the values from a ratio table as points on the coordinate plane.

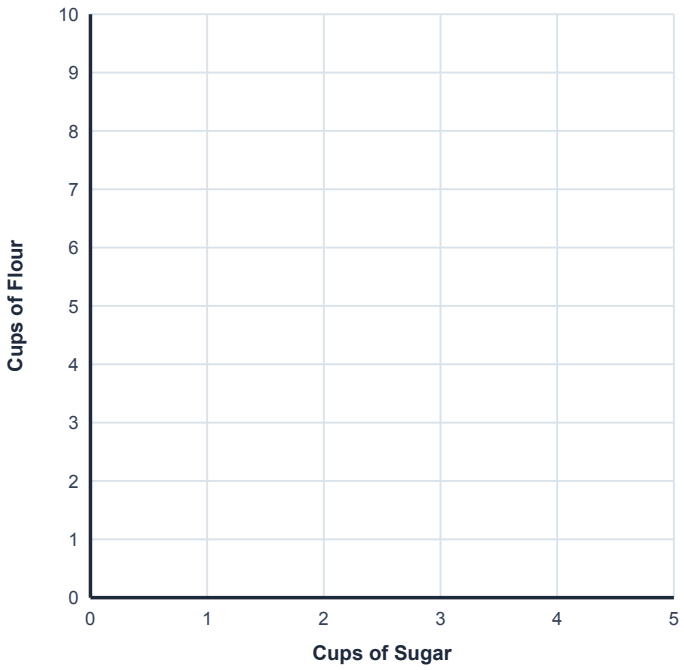
WHY THIS MATTERS The organizers of a soccer league ordered 6 of the bags shown to hold soccer balls. Each bag holds the same number of soccer balls. How many soccer balls will the bags hold?

PART 1 Understand the idea

models and meaning

1 **Plot and label each point on the grid.**

A recipe uses 1 cup of sugar for every 2 cups of flour. Plot the ordered pairs: (1, 2), (2, 4), (3, 6).



HINT Each pair is (sugar, flour): the first number tells you how far RIGHT, the second how far UP.

SHOW YOUR THINKING

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A ratio table shows (2, 6), (4, 12), and (6, 18). Which ordered pair comes next if the pattern continues?

(A) (7, 20)

(B) (8, 24)

(C) (8, 22)

(D) (10, 24)

HINT Watch each coordinate separately: x goes 2, 4, 6, … and y goes 6, 12, 18, …

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

When you graph equivalent ratios, the points will always form what shape?

(A) A curved line

(B) A straight line through the origin

(C) A zigzag

(D) A circle

HINT Think about what happened when you plotted equivalent-ratio points earlier in this lesson.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A ratio table shows (2, 8), (3, 12), and (5, 20). What is the y-value when x = 7?

(A) 21
(B) 24
(C) 28
(D) 35

HINT Compare each pair: 2 → 8, 3 → 12, 5 → 20. What is done to x to get y every time?

My work — one step per line	What I did to both sides

1. **Look** *What is done to the variable?*
2. **Undo** *Do the opposite — both sides.*
3. **Simplify** *One side at a time.*
4. **Check** *Put it back in.*

5 **Circle the letter of the best answer. Show how you know.**

Which point would NOT be on the graph of the ratio 1:5?

(A) (2, 10)
(B) (3, 10)
(C) (4, 20)
(D) (5, 25)

HINT On the 1:5 graph, every y-value is exactly 5 times its x-value.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Level 1 — Spot the Common Mistake

1. **Read the ratio** A recipe uses 1 cup of sugar for every 2 cups of flour. Plot the point for 3 cups of sugar.
2. **Find the flour** 3 cups of sugar means 3 + 2 = 5 cups of flour.
3. **Write the ordered pair** (3, 5)

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Look at Step 2. The student added 2 to the sugar instead of multiplying by 2.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

What pattern do the plotted points make? How does the graph show the ratio of 1 sundae to 2 ounces of sauce?

SENTENCE STARTER Every time the sundaes go up by ___, the ounces go up by ___, so the points line up in a ___ starting from ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 3.4

Determine Equivalent Ratios Using Graphs

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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LEARNING TARGET I can graph the values from a ratio table as points on the coordinate plane.

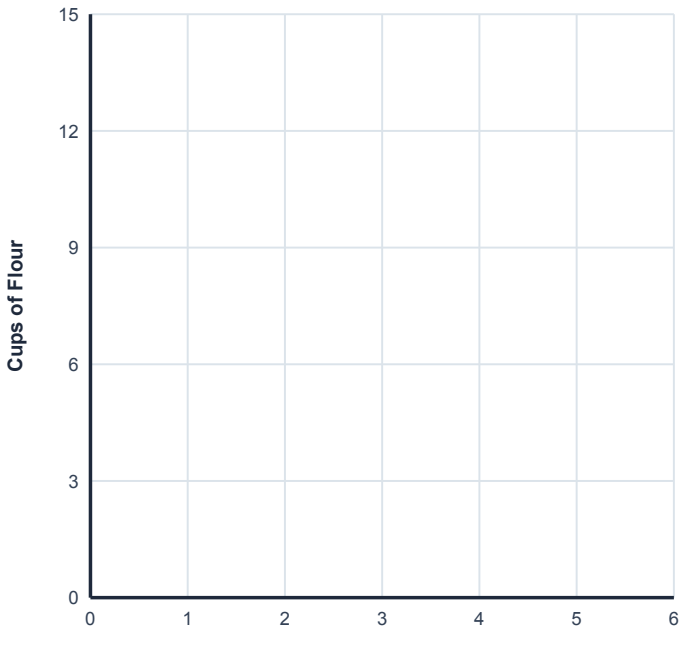
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PART 1 Understand the idea

models and meaning

1 **Plot and label each point on the grid.**

Chef Reyes uses 3 cups of flour for every 1 cup of sugar. Plot the ordered pairs from the ratio table: (1,3), (2,6), (3,9), (4,12).



SHOW YOUR THINKING

2 **Complete the table. Show how you found each value.**

Use the graph pattern y = 5x to complete the ratio table, then identify what the ordered pairs would be.

x	y	Ordered Pair
1	5	(1, 5)
2		
4		
6		

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Points from a ratio table are plotted at (1, 4), (2, 8), and (3, 12). What is the ratio of y to x for each point?

(A) 1:4
(B) 2:8
(C) 3:4
(D) 4:1

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Two students graph ratio tables. Student A plots (1,2), (2,4), (3,6). Student B plots (1,3), (2,6), (3,9). Whose line is steeper?

(A) Student A
(B) Same steepness
(C) Cannot tell
(D) Student B

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 **Find the mistake. Explain it, then show the correct work.**

On Level — Spot the Common Mistake

1. **Read the ratio** A recipe uses 4 cups of water for every 1 cup of rice (rule: y = 4x). A student is plotting points for the line.
2. **List the points** (1, 4), (2, 8), (3, 12)
3. **Add another point** The student also plots (4, 14) and says it belongs on the same line.
4. **Conclusion** All four points form a straight line.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

What pattern do the plotted points make? How does the graph show the ratio of 1 sundae to 2 ounces of sauce?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 3.4

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Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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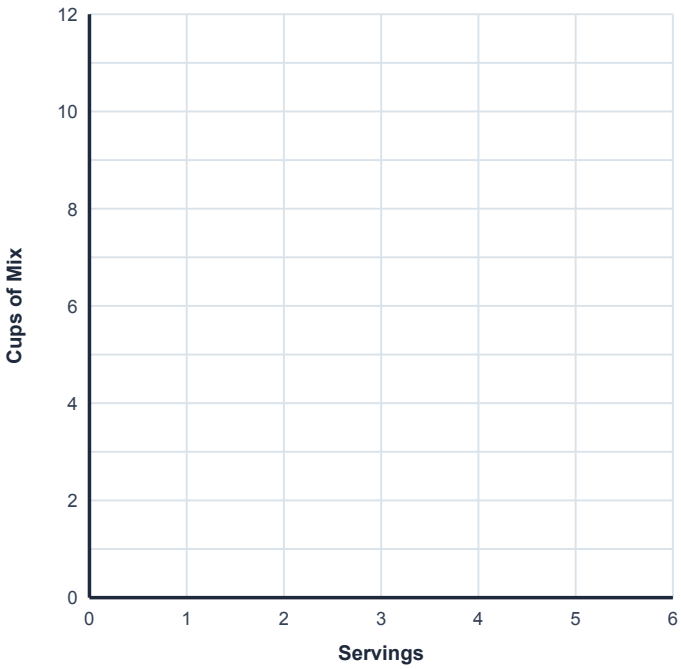
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PART 1 Understand the idea

models and meaning

1 **Plot and label each point on the grid.**

Two recipes are graphed: Recipe A uses 2 cups of mix per 1 serving, and Recipe B uses 3 cups of mix per 2 servings. Plot both sets of points and determine which recipe uses more mix per serving.



SHOW YOUR THINKING

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find Jada's Mistake

1. **Read the ratio table** Cookies: 2, 4, 6 | Cups of milk: 1, 2, 3
2. **Write ordered pairs** (1, 2), (2, 4), (3, 6)
3. **Plot the points** Plotted (1,2), (2,4), (3,6) on the grid
4. **Conclusion** The x-axis is cookies and y-axis is cups of milk

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

A student plots the points (1, 3), (2, 6), (3, 9), and (4, 15) from a ratio table. She says they form a straight line because they go up. Is she correct? Explain how to check.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

What pattern do the plotted points make? How does the graph show the ratio of 1 sundae to 2 ounces of sauce?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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